Q-1) Solve the I.V.P

$(D^3 - 4D)y = 10\cos x + 5\sin x$

$y(0) = 3 \quad y'(0) = -2 \quad y''(0) = -1$

Soln) Solve the homogeneous part

$y''' - 4y' = 0$

$r^3 - 4r = 0$ (Characteristic Eqn)

$r(r^2 - 4) = 0$

$r = 0, 2, -2$

So,

$y_h = C_1 + C_2 e^{2x} + C_3 e^{-2x}$

RHS $\quad 10\cos x + 5\sin x$

assume $y_p = A\cos x + B\sin x$

$y_p' = -A\sin x + B\cos x$

$y_p'' = -A\cos x - B\sin x$

$y_p''' = +A\sin x - B\cos x$

substituting in D.E

$= (A\sin x - B\cos x) - 4(-A\sin x + B\cos x)$

$= (A + 4A)\sin x + (-B - 4B)\cos x$

$= 5A\sin x - 5B\cos x$

comparing with RHS $\quad 10\cos x + 5\sin x$

$A = 1 \qquad B = -2$

$y_p = \cos x - 2\sin x$

General solution

$y = y_h + y_p$

②

$$y = C_1 + C_2 e^{2x} + C_3 e^{-2x} + \cos x - 2\sin x$$

applying initial conditions

$y(0) = 3$

$C_1 + C_2 + C_3 + 1 = 3$

$C_1 + C_2 + C_3 = 2$ — (1)

$y'(0) = -2$

$y'(x) = 2C_2 e^{2x} - 2C_3 e^{-2x} - \sin x - 2\cos x$

$\Rightarrow 2C_2 - 2C_3 - 2 = -2$

$\Rightarrow C_2 = C_3$ — (2)

$y''(0) = -1$

$\Rightarrow y'' = 4C_2 e^{2x} + 4C_3 e^{-2x} - \cos x + 2\sin x$

$y''(0) = -1$

$\Rightarrow 4C_2 + 4C_3 - 1 = -1$

$\Rightarrow C_2 + C_3 = 0$ — (3)

$\Rightarrow C_2 = C_3 = 0$ & $C_1 = 2$

$$\boxed{y = 2 + \cos x - 2\sin x}$$ (Ans)

Q-2) Solve the ODE

$$x^3 y''' + 2x^2 y'' - xy' + y = \frac{1}{x^2}$$

Soln)

This is Cauchy Euler Eqn

put $y = x^m$

Substitute into the Homogeneous Eqn

$$x^3 [m(m-1)(m-2) x^{m-3}]$$
$$+ 2x^2 [m(m-1) x^{m-2}] - x[m x^{m-1}] + x^m = 0$$

So the Auxillary Eq is

$$m(m-1)(m-2) + 2m(m-1) - m + 1 = 0$$

$$m^3 - m^2 - m + 1 = 0$$

roots $\quad m = 1, 1, -1$

$$\therefore \; y_h = c_1 x + c_2 x \ln x + c_3 x^{-1}$$

For particular solution

$y_p = A x^{-2}$ assume

$y_p' = -2Ax^{-3}$ $\quad y_p'' = 6Ax^{-4}$ $\quad y_p''' = -24Ax^{-5}$

Substitute into LHS

$$= x^3(-24Ax^{-5}) + 2x^2(6Ax^{-4}) - x(-2Ax^{-3})$$
$$+ Ax^{-2}$$

$$= -9Ax^{-2}$$

~~Equating~~ $\quad$ L.H.S = R.H.S

$$\Rightarrow -9Ax^{-2} = x^{-2}$$

$$-9A = 1$$

$$A = -\frac{1}{9}$$

$$\Rightarrow y_p = -\frac{1}{9x^2}$$

General Solution

$$y = y_h + y_p$$

$$y = c_1 x + c_2 x \ln x + \frac{c_3}{x} - \frac{1}{9x^2}$$

where c_1, c_2 and c_3 are arbitrary constants.

(4)

Q-3) Assume that all the zeros of the polynomial $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ have negative real parts. If $u(t)$ is any solution to the ODE

$$a_n \cdot \frac{d^n u}{dt^n} + a_{n-1}\frac{d^{n-1}u}{dt^n} + \cdots + a_1\frac{du}{dt} + a_0 u = 0$$

then $\lim_{t\to\infty} u(t)$ is equal to ________

Soln) Let $\alpha \pm i\beta$, $\alpha < 0$ be roots of $a_n x^n + a_{n-1}x^n + \cdots + a_1 x + a_0$ then one part $u(t)$ is

$$e^{\alpha t}(c_1\cos\beta t + c_2 \sin\beta t)$$

and when $t \to \infty$

$\cancel{u(t)}$ $e^{\alpha t}(c_1\cos\beta t + c_2\sin\beta t) \to 0$ as $\alpha < 0$

and $c_1 \cos\beta t + c_2 \sin\beta t$ is bounded

This part is true for each part of $u(t)$

So $\lim_{t\to\infty} u(t) = 0$ (Ans)

Q-4) The boundary value problem

$$x^2 y'' - 2xy' + 2y = 0, \quad 1 \le x \le 2$$

$$y(1) - y'(1) = 1$$

$$y(2) - ky'(2) = 4$$

has infinitly many distinct solution when k is equal to ________

Soln) Cauchy Euler Eq^n

put $y = x^m$

$\Rightarrow \cancel{[m(m-1)x^{m-2}] - 2mx^{m-1} + 2x^m = 0}$

$\Rightarrow x^2[m(m-1)x^{m-2}] - 2x[mx^{m-1}] + 2x^m = 0$

$\Rightarrow x^m[m(m-1) - 2m + 2] = 0$

$\Rightarrow m^2 - 3m + 2 = 0$

$m = 1, 2$

$y = C_1 x + C_2 x^2$ is the general solution

where C_1 & C_2 are arbitrary constant

$y(x) = C_1 x + C_2 x^2$

$y'(x) = C_1 + 2C_2 x$

$y(1) - y'(1) = 1$ (Applying Intial condition)

$\Rightarrow C_1 + C_2 - (C_1 + 2C_2) = 1$

$\Rightarrow C_2 = -1$

and

$y(2) - ky'(2) = 4$

$\Rightarrow 2C_1 - 4 - k(C_1 - 4) = 4$

$\Rightarrow (2-k)C_1 + 4(k-2) = 0$

if $k = 2$ then $0 = 0$

So infinitely many solution

$\boxed{k = 2}$ (Ans)

Q - 5) Prey Predator Model

Two species interact in an ecosystem: a prey population $x(t)$ and a predator population $y(t)$

Assume:

1) In the absence of predators, the prey grows at a rate proportional to its population.
2) In the absence of prey, the predator population decays at a rate proportion to its population.
3) The interaction between predator and prey is proportion to the product of their populations.

Using these assumptions, formulate the system of differential equation describing the dynamics of populations

Soln $x(t)$ = population of prey at time t
$y(t)$ = population of predator at time t.
t = time

I) Prey Growth Without Predators

$$\frac{dx}{dt} = ax$$

$a > 0$ is the natural growth rate of prey

II) Predator Decay Without Prey

$$\frac{dy}{dt} = -cy$$

$c > 0$ is the natural death rate of predators

III) Interaction between predator and prey

when both species interact:

1) The rate of encounters is proportion to xy.
2) Prey decreases due to predation
3) Predator increases due to consuming prey.

So we assume

* Loss of prey due to predation is proportional to xy
* Gain of predator due to predation is proportion to xy.

So prey equation

$$\frac{dx}{dt} = ax - bxy$$

$b > 0 \rightarrow$ predation rate coefficient

predator equation

$$\frac{dx}{dt} = -cy + dxy$$

$d > 0 \rightarrow$ predator growth rate per prey consumed

Final System

$$\boxed{\begin{aligned} \frac{dx}{dt} &= ax - bxy \\ \frac{dy}{dt} &= -cy + dxy \end{aligned}}$$

This is famous Lotka Valterra System